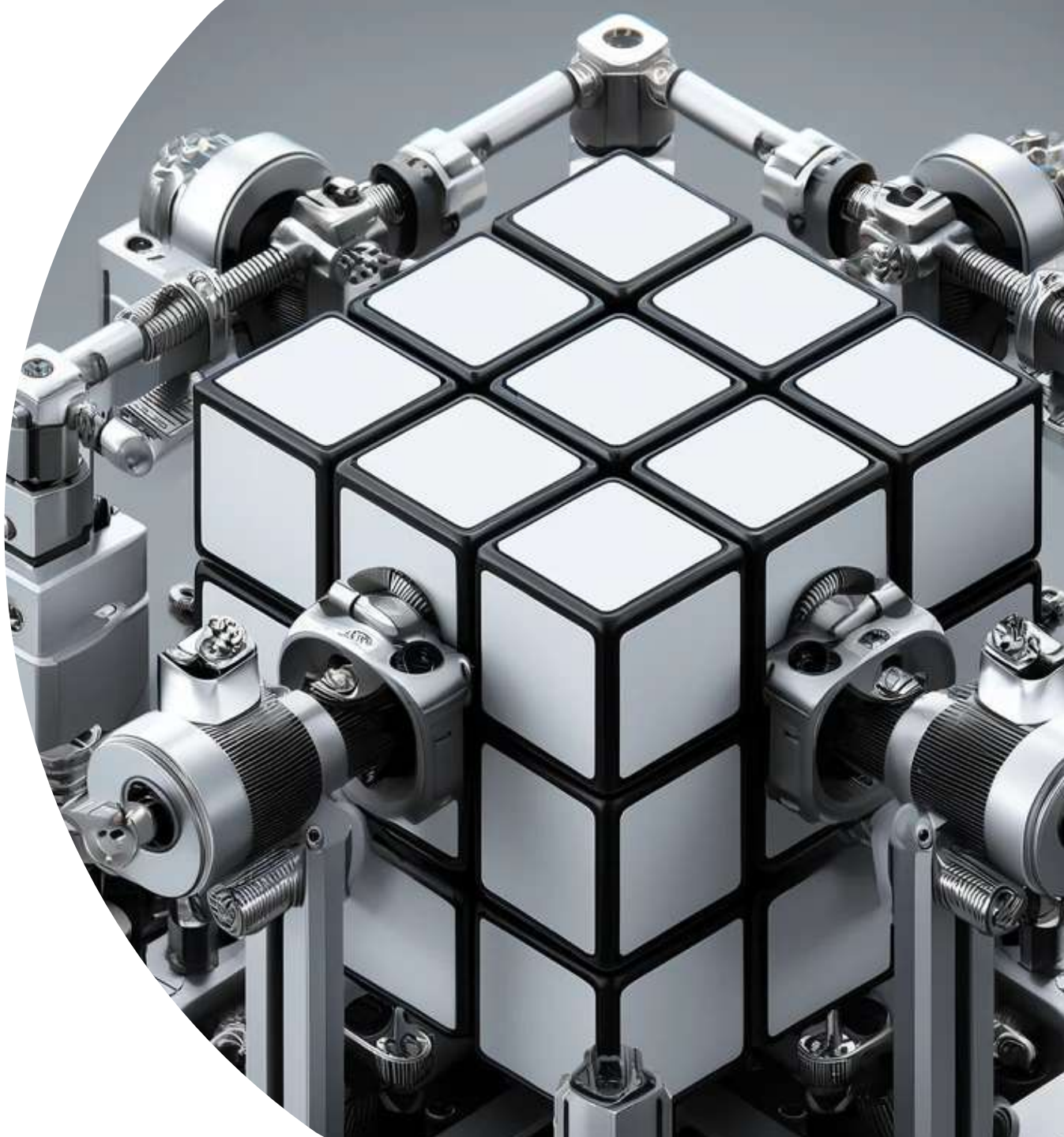
Un robot résolvant le Rubik's Cube

TIPE 2024 *:* **Jeux Et Sports**



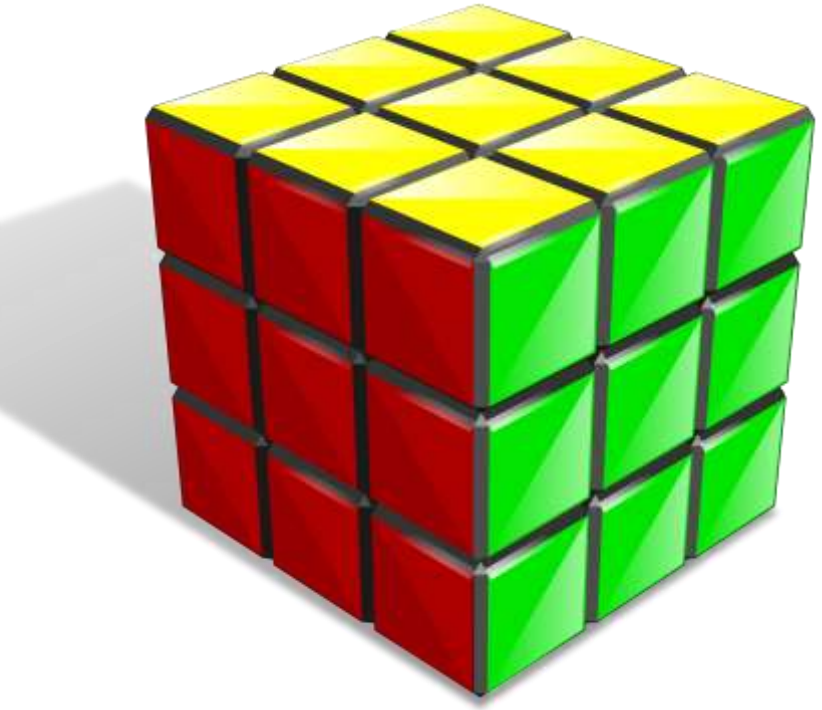
Réalisé par : BRAHIM ES SELLAMI

Ouverture

Figure 1 : Compétition de speedcubing

Figure 2 : Un Australien remporte le championnat du monde de Rubik's Cube

PLAN



- Description de système
- Identification de la problématique
- Présentation des composants électronique
- Solutions d’améliorations proposées
- Fonctionnement du système
- Conclusion

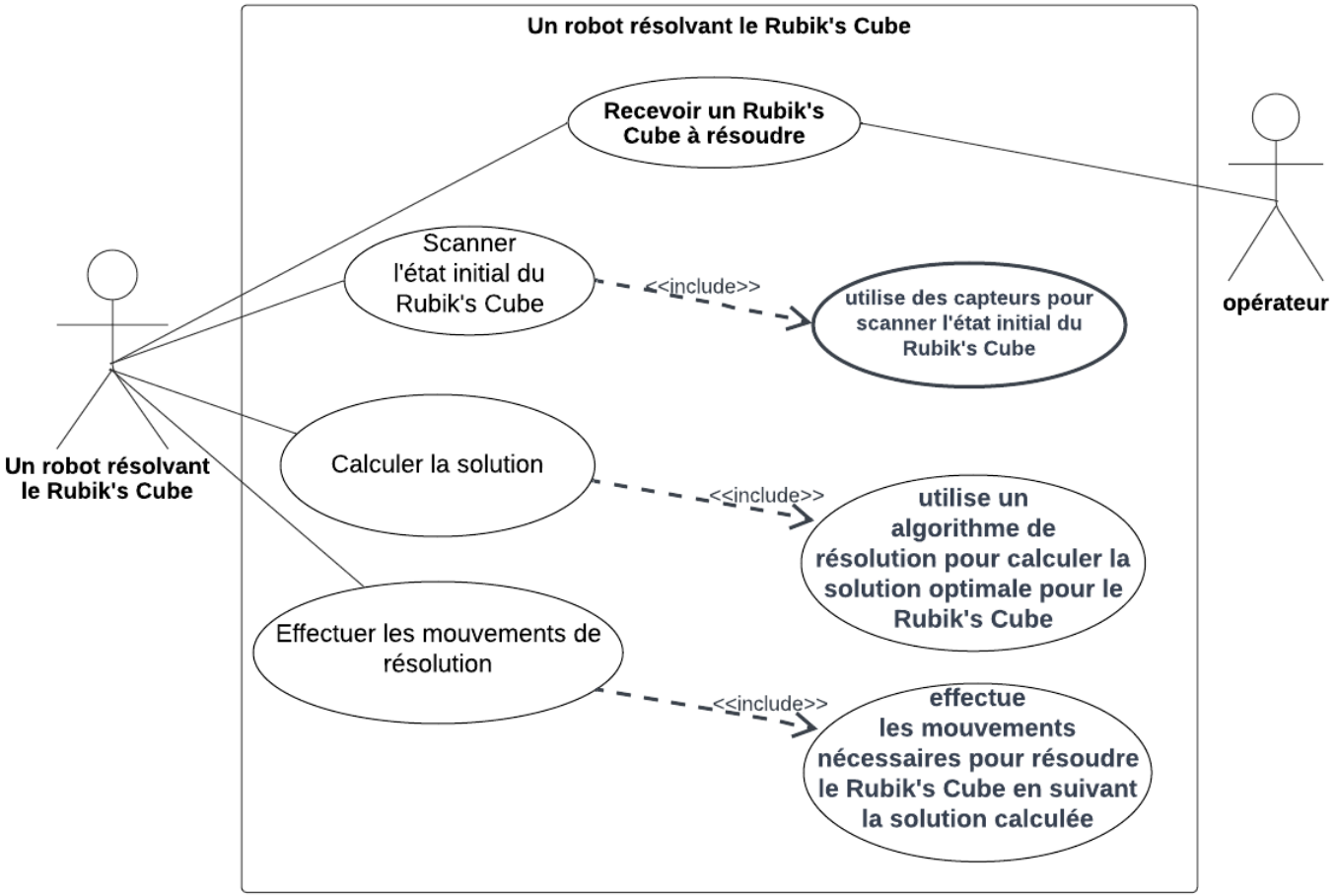


Figure 3 : Diagramme de cas d'utilisation d'un robot résolvant le Rubik's Cube

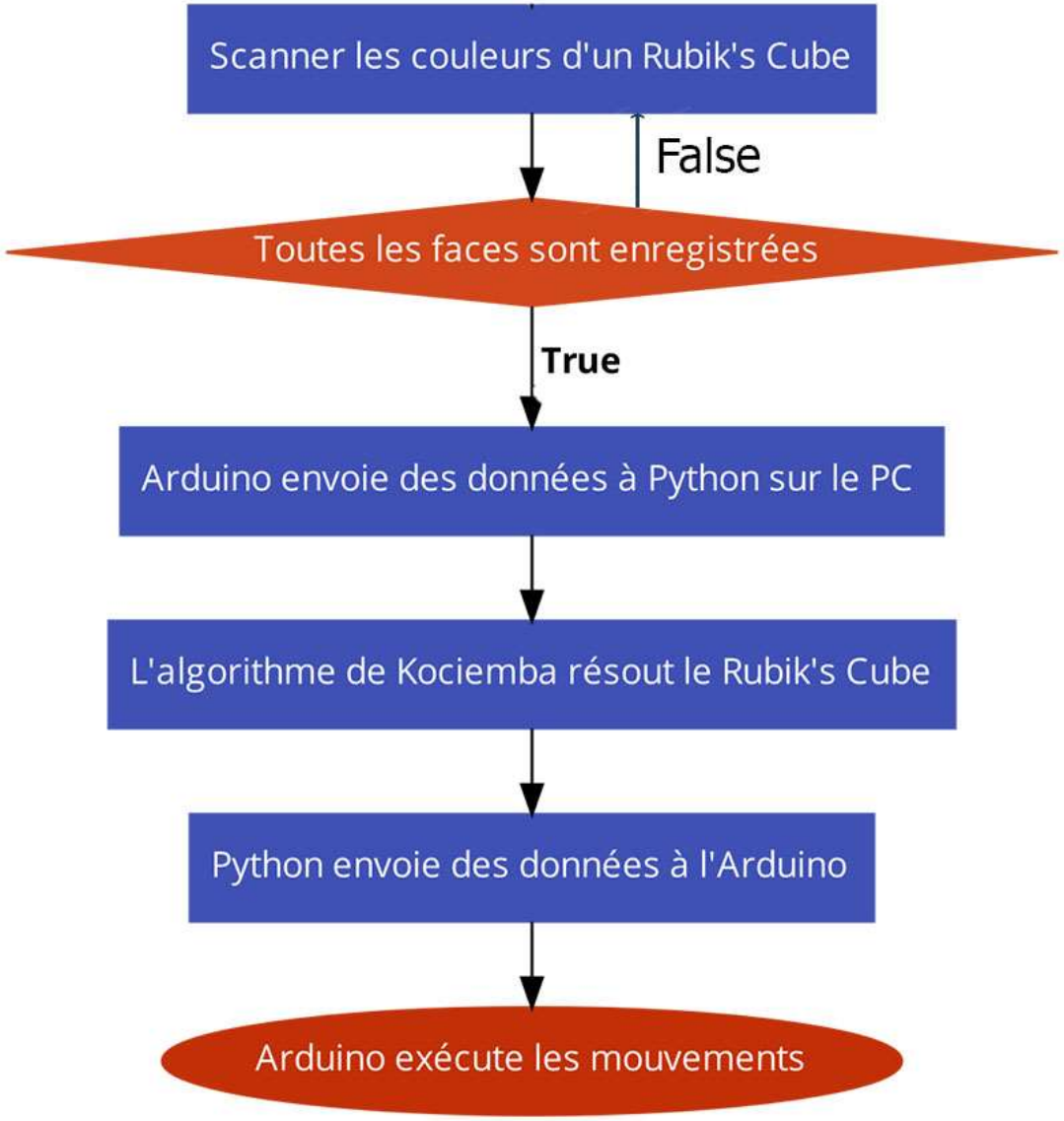
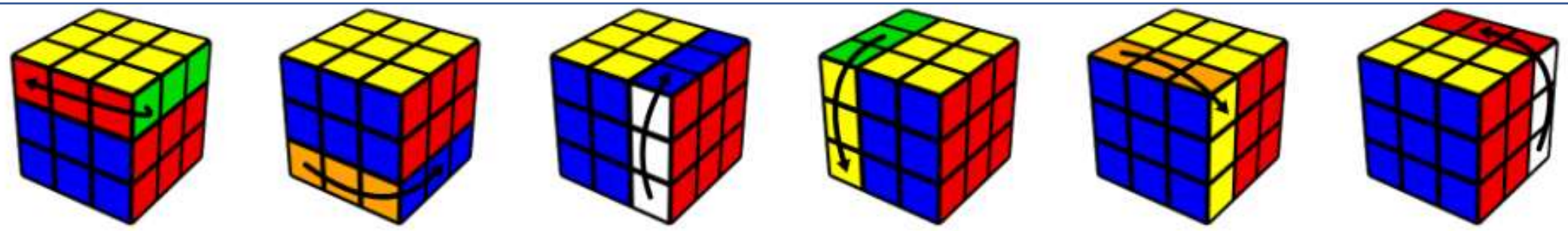
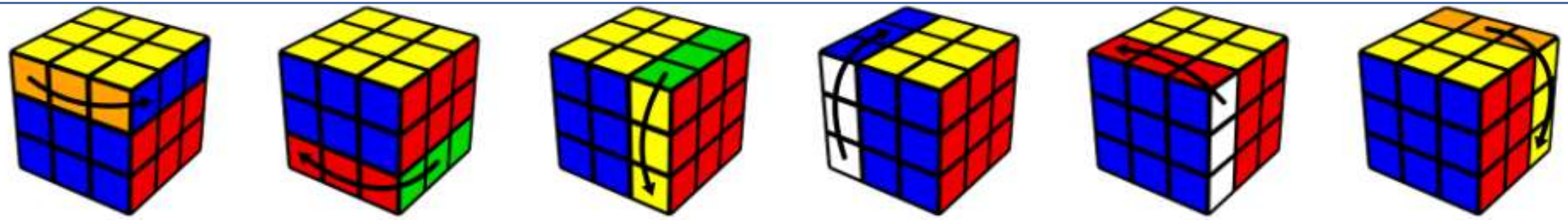


Figure 4 : Organigramme du robot lors de la résolution

Les mouvements de base sont Haut, Bas, Droite, Gauche, Avant, Arrière.



U D R L F B



U' D' R' L' F' B'

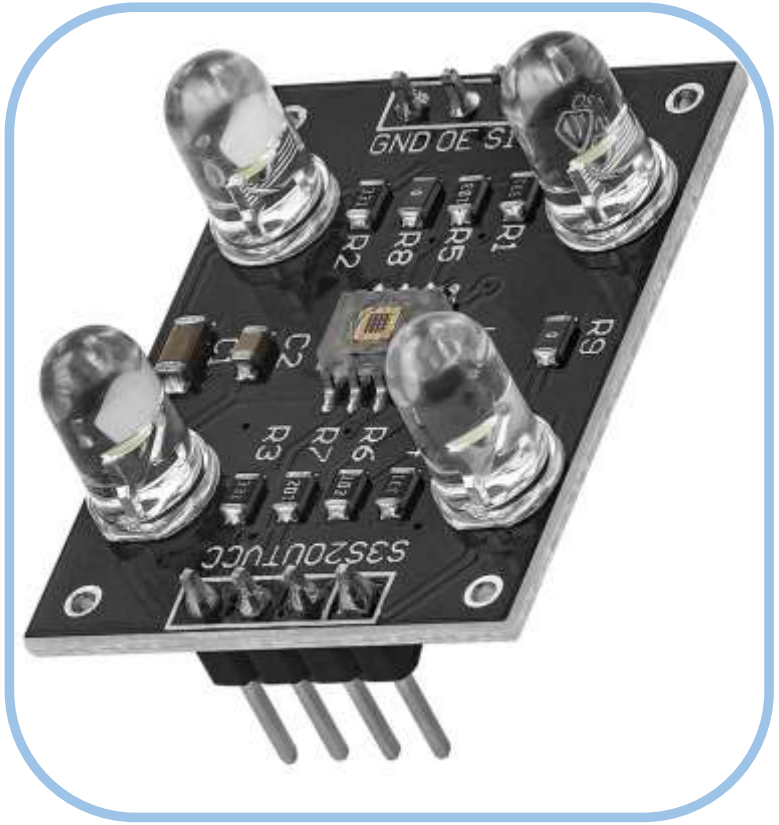
Problématique retenue :

Prenons en considération l’existence d’un ensemble de méthode de résolution robotique mon sujet met l’accent sur la question suivante : Comment étalonner le capteur de couleurs pour mieux commander les actionneurs afin de résoudre un Rubik's Cube ?

Les capteur utiliser avec Arduino :

capteur de couleurs TCS230:

Alimentation	2,7 V à 5,5 V
Taille	28,4 x 28,4 mm
Interface	TTL numérique



- Conversion haute résolution de l'intensité lumineuse en fréquence
- Couleur programmable et fréquence de sortie pleine échelle
- Communique directement au microcontrôleur

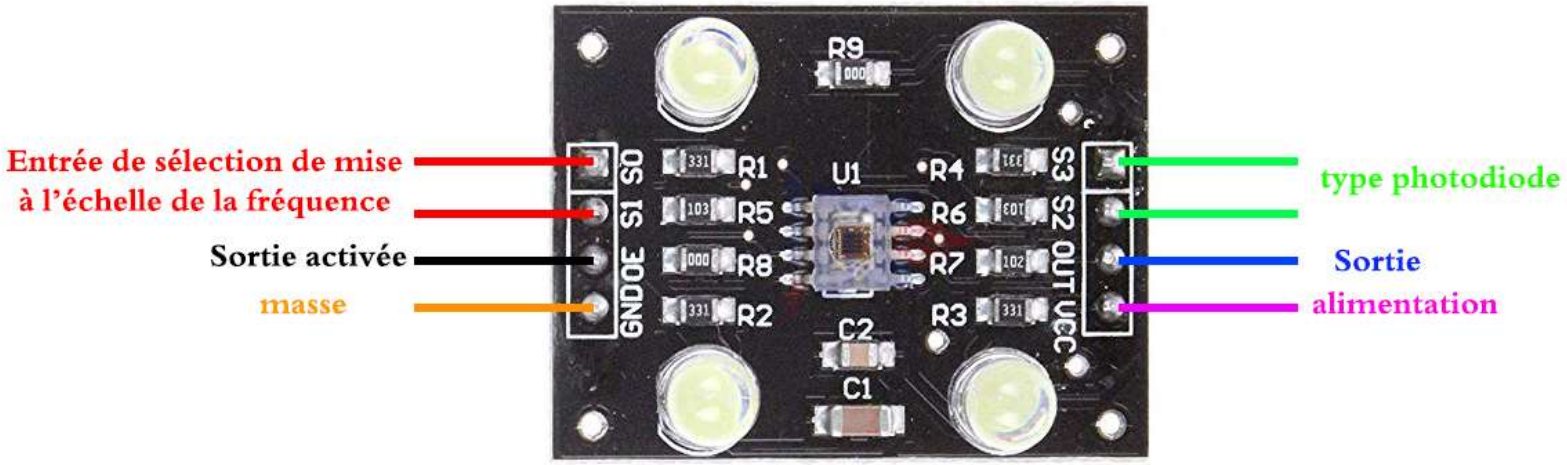
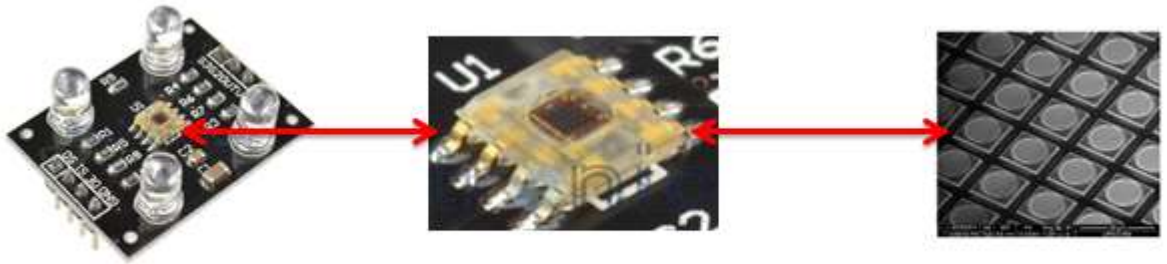


Figure 1 : capteur de couleurs TCS230



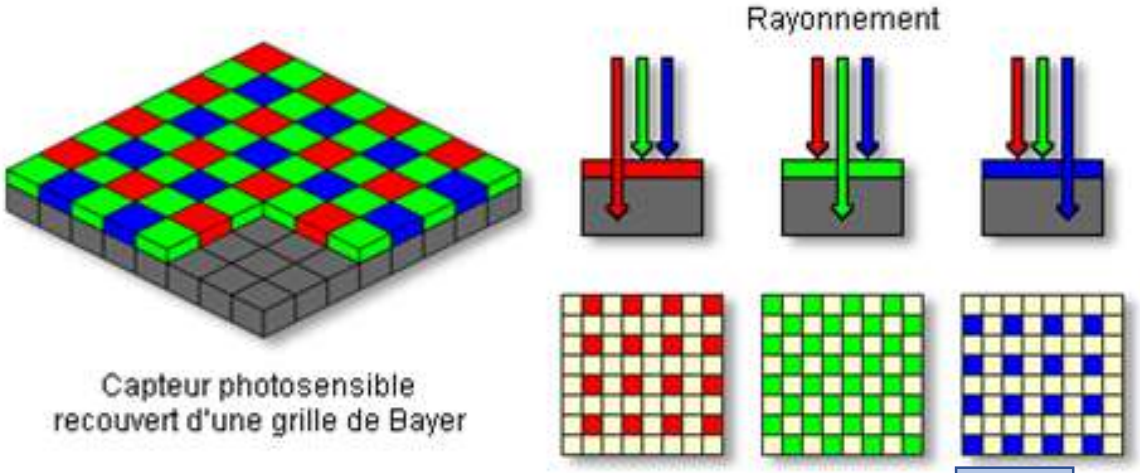
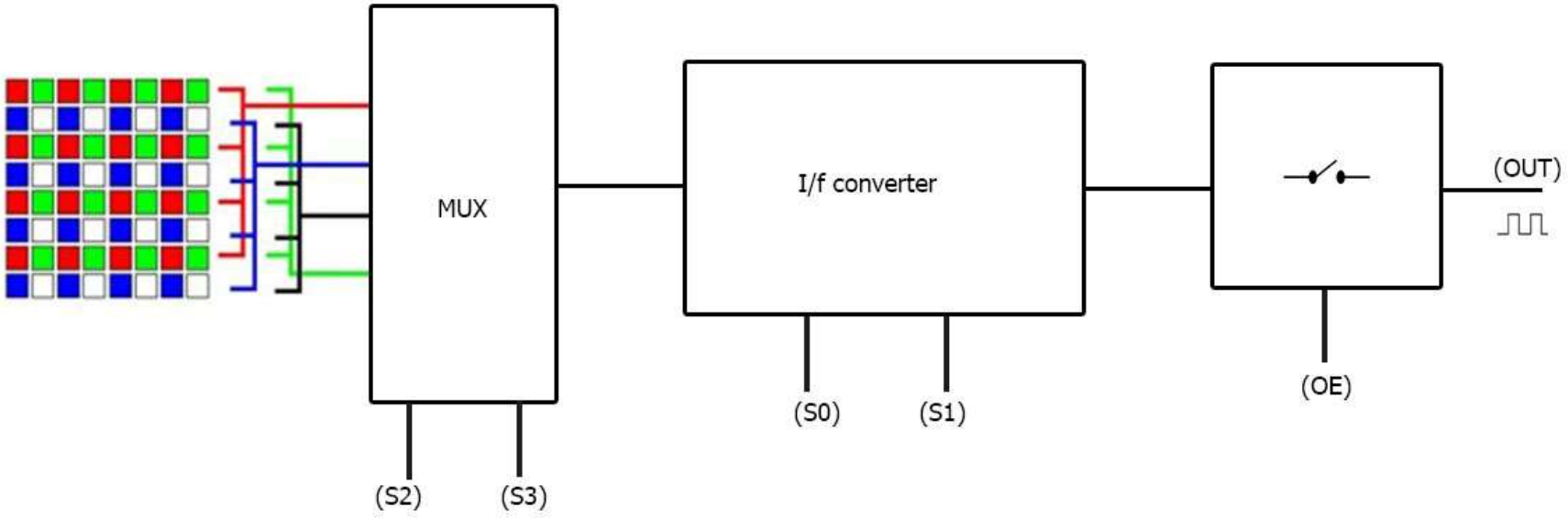


Figure 5 : Zoom sur la matrice de photodétecteurs

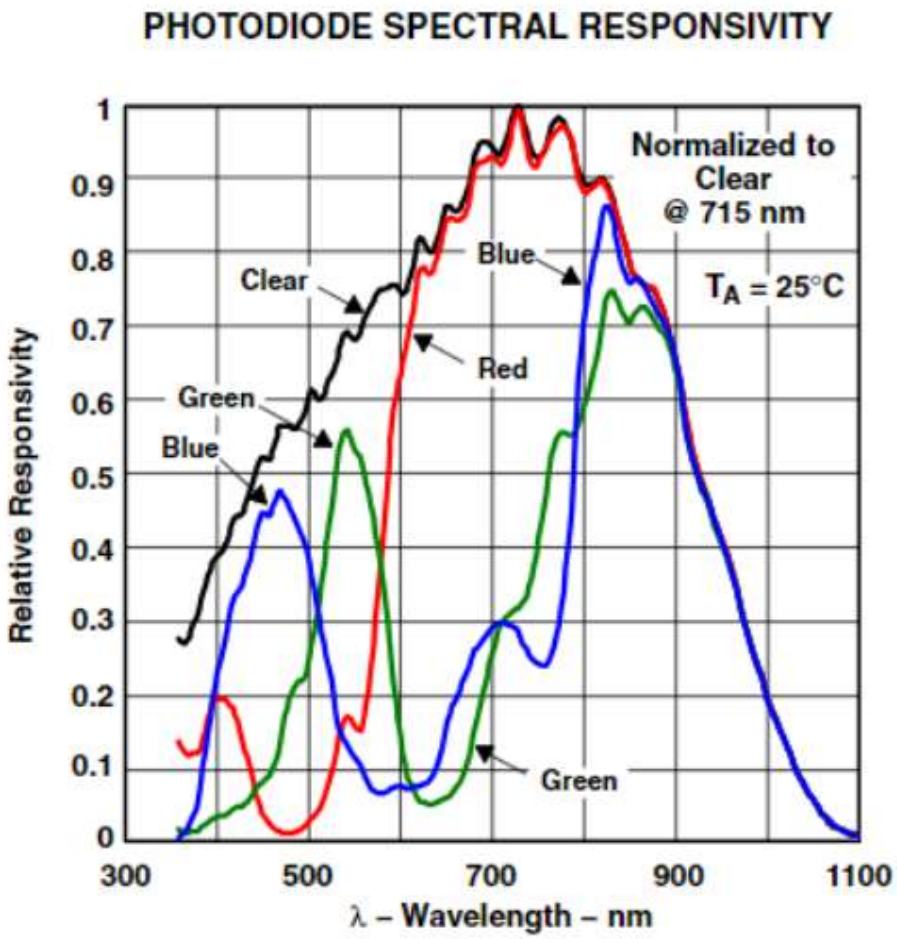
Composition du Robot

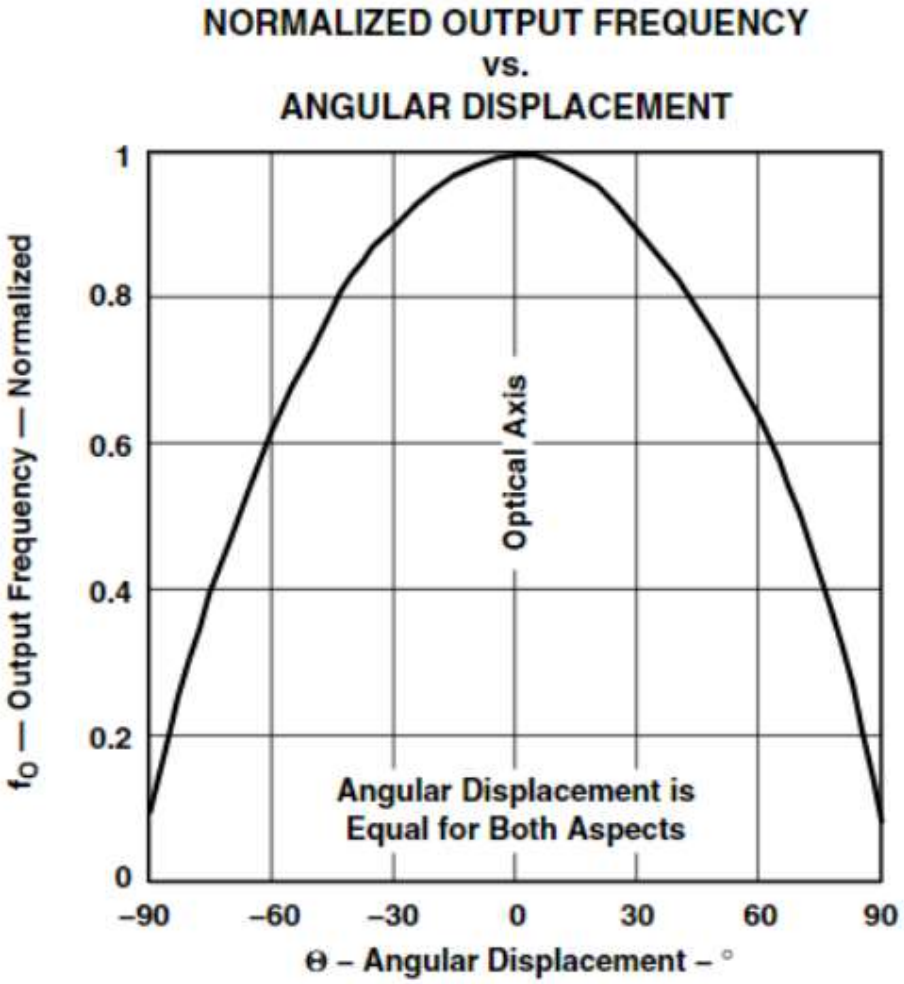


S2	S3	PHOTODIODE TYPE
L	L	Red
L	H	Blue
H	L	Clear (no filter)
H	H	Green

S0	S1	OUTPUT FREQUENCY SCALING (f_o)
L	L	Power down
L	H	2%
H	L	20%
H	H	100%

Composition du Robot





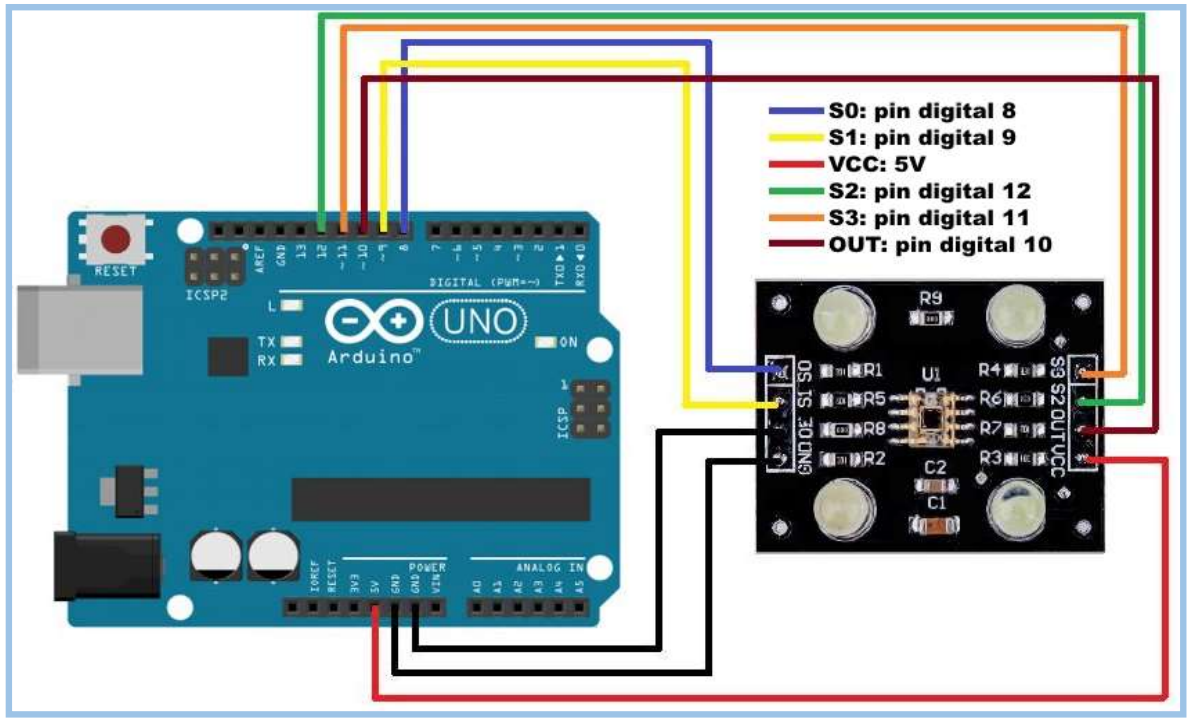


Figure 6 :Schéma éclectique du montage

```
#define S0 4
#define S1 5
#define S2 6
#define S3 7
#define sensorOut 8

int redPW = 0;
int greenPW = 0;
int bluePW = 0;

void setup() {
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);

  digitalWrite(S0,HIGH);
  digitalWrite(S1,LOW);

  pinMode(sensorOut, INPUT);
  Serial.begin(9600);
}
```

Figure 7 :Initialisation d'un capteur de couleur pour la détection RGB

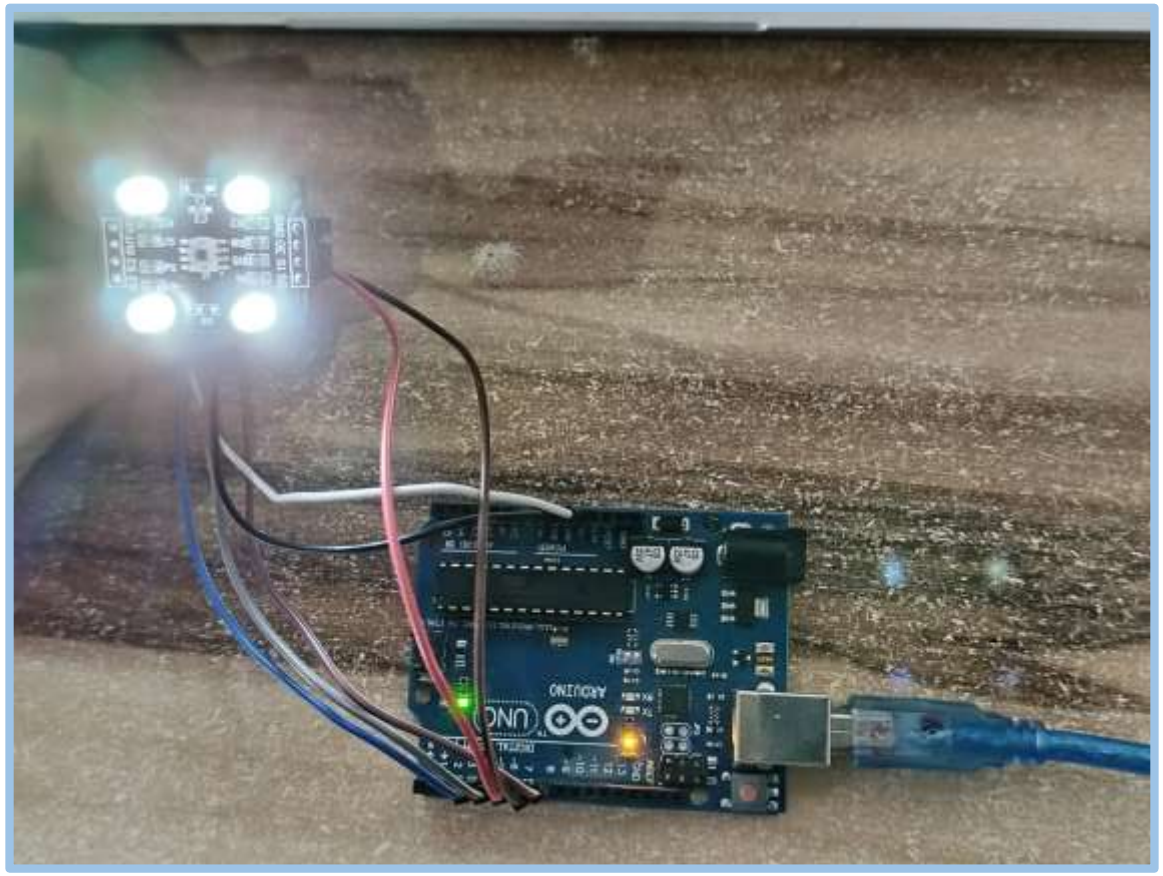


Figure 8 :Branchement le capteur de couleur rgb Arduino

```
void loop() {
  digitalWrite(S2, LOW);
  digitalWrite(S3, LOW);
  redFrequency = pulseIn(sensorOut, LOW);
  digitalWrite(S2, HIGH);
  digitalWrite(S3, HIGH);
  greenFrequency = pulseIn(sensorOut, LOW);
  digitalWrite(S2, LOW);
  digitalWrite(S3, HIGH);
  blueFrequency = pulseIn(sensorOut, LOW);

  Serial.print("R = ");
  Serial.print(redFrequency);
  Serial.print(" G = ");
  Serial.print(greenFrequency);
  Serial.print(" B = ");
  Serial.println(blueFrequency);

  delay(1000);
```

Figure 9 :Lecture des niveaux de puissance des couleurs RVB

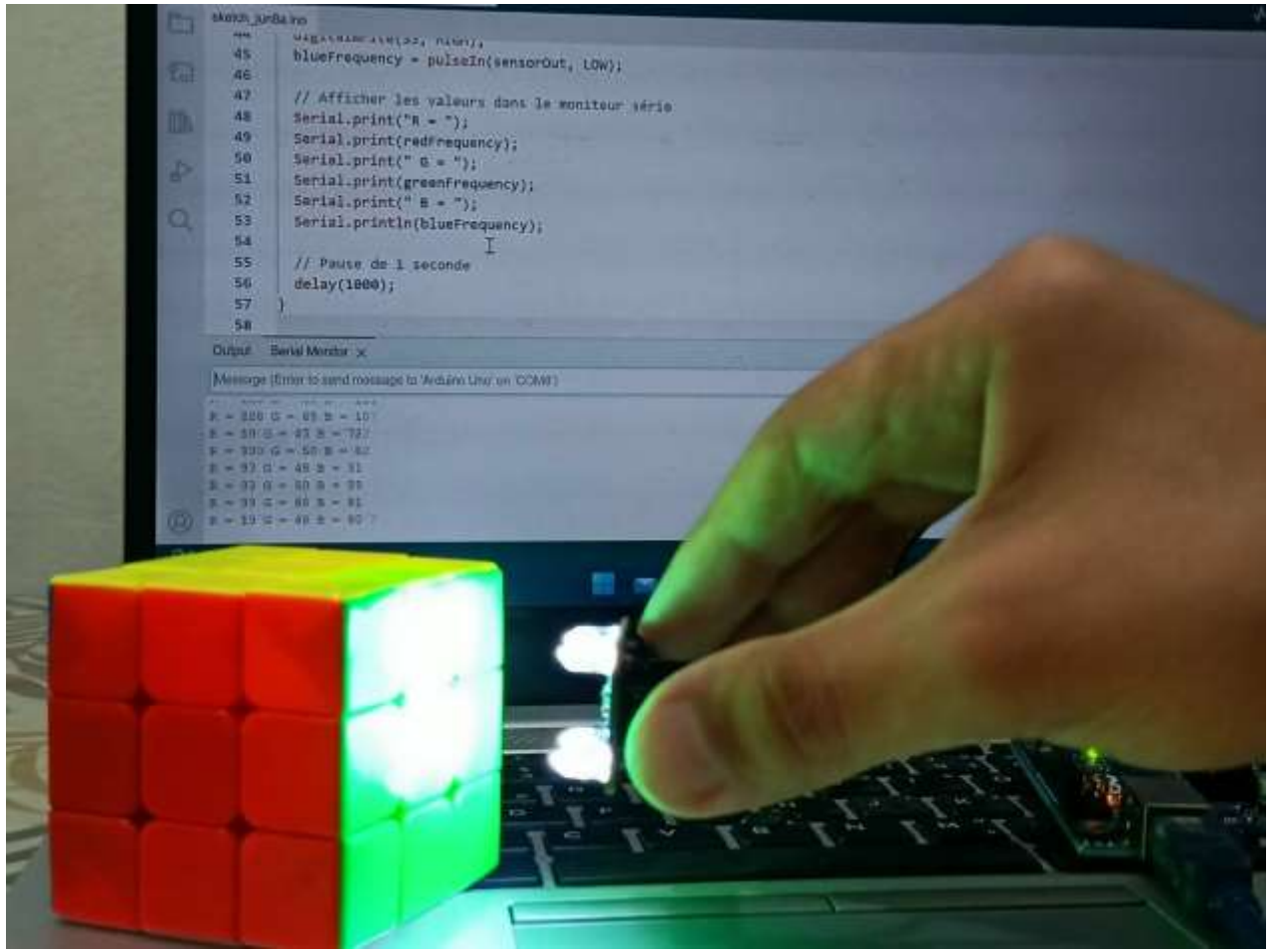


Figure 10 :la détection de couleur en conditions réelles

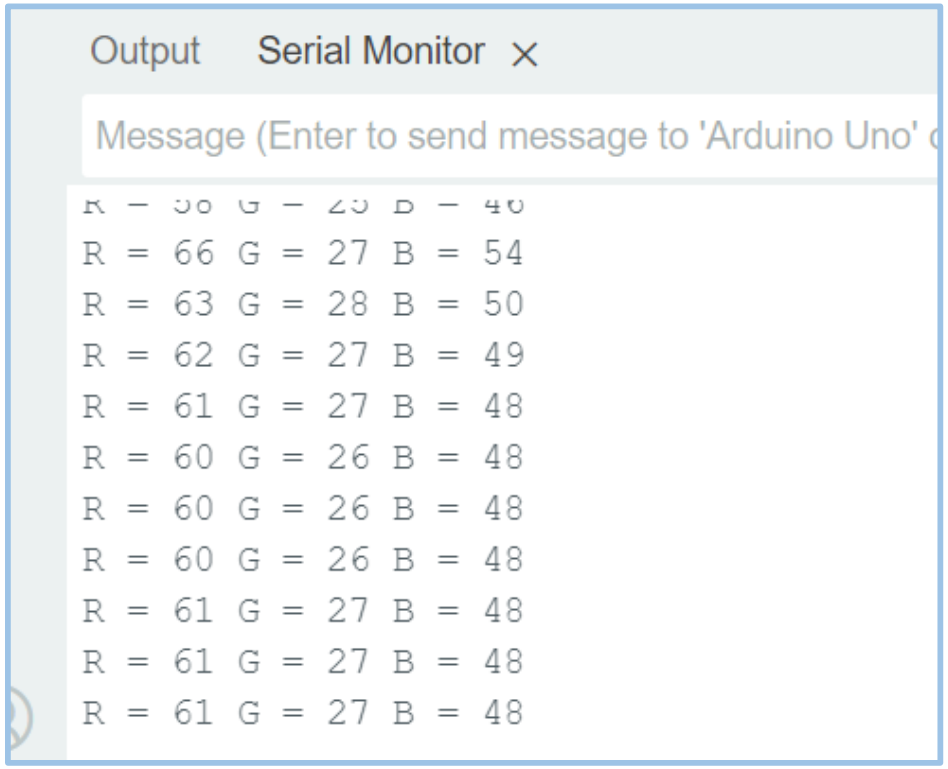


Figure 11 :résultat

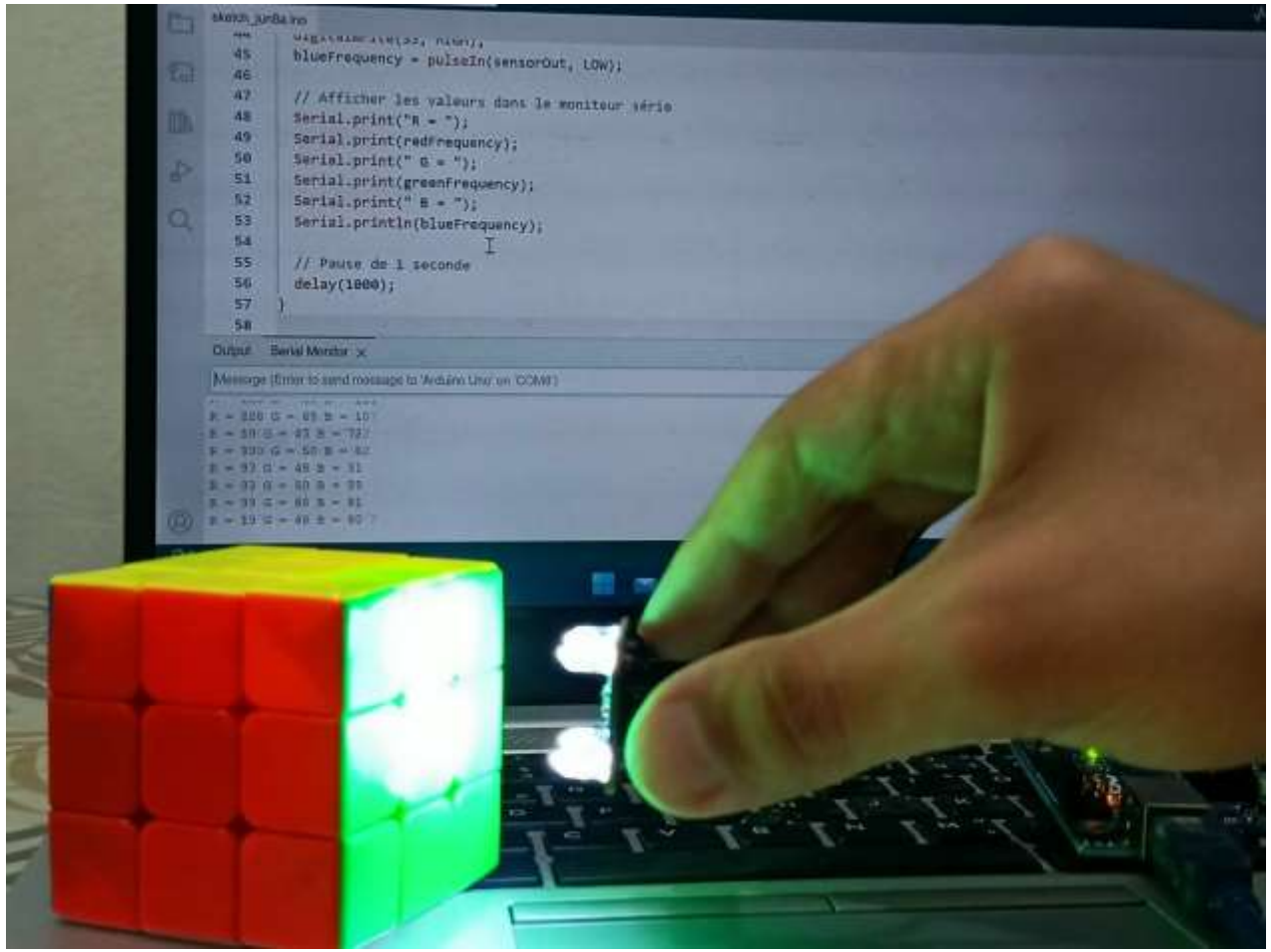


Figure 9 :la détection de couleur en conditions réelles

Composition du Robot

R,G,B∈[200,255]

R∈[200,255],G∈[100,200],B∈[0,100]

R∈[200,255],G∈[200,255],B∈[0,100]

R∈[0,100],G∈[0,100],B∈[150,255]

R∈[150,255],G∈[0,100],B∈[0,100]

R∈[0,100],G∈[150,255],B∈[0,100]

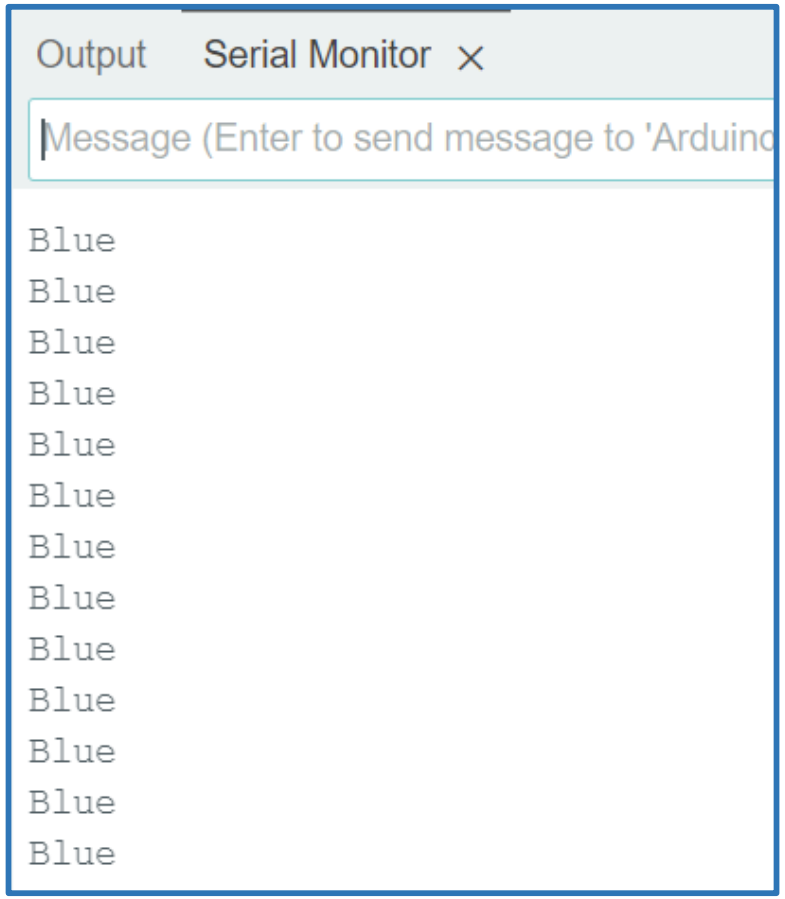
```
void loop() {
  // Execute the GetColors function to get the value of each RGB color
  GetColors();

  // Compare the color values to identify the color
  if (Red <= 15 && Green <= 15 && Blue <= 15)
    Serial.println("White");
  else if (Blue > 20 && Red < 10 && Green < 10)
    Serial.println("Blue");
  else if (Green > 20 && Red < 10 && Blue < 10)
    Serial.println("Green");
  else if (Red > 20 && Green > 5 && Blue < 10)
    Serial.println("Orange");
  else if (Red > 20 && Green > 20 && Blue < 10)
    Serial.println("Yellow");
  else if (Red > 20 && Green < 10 && Blue < 10)
    Serial.println("Red");
  else
    Serial.println("Unknown");

  delay(2000);  // 2s delay, you can modify if you want
}
```

Figure 12 :Stabilisation des Couleurs du Rubik's Cube

Figure 13 :détection du couleur bleu



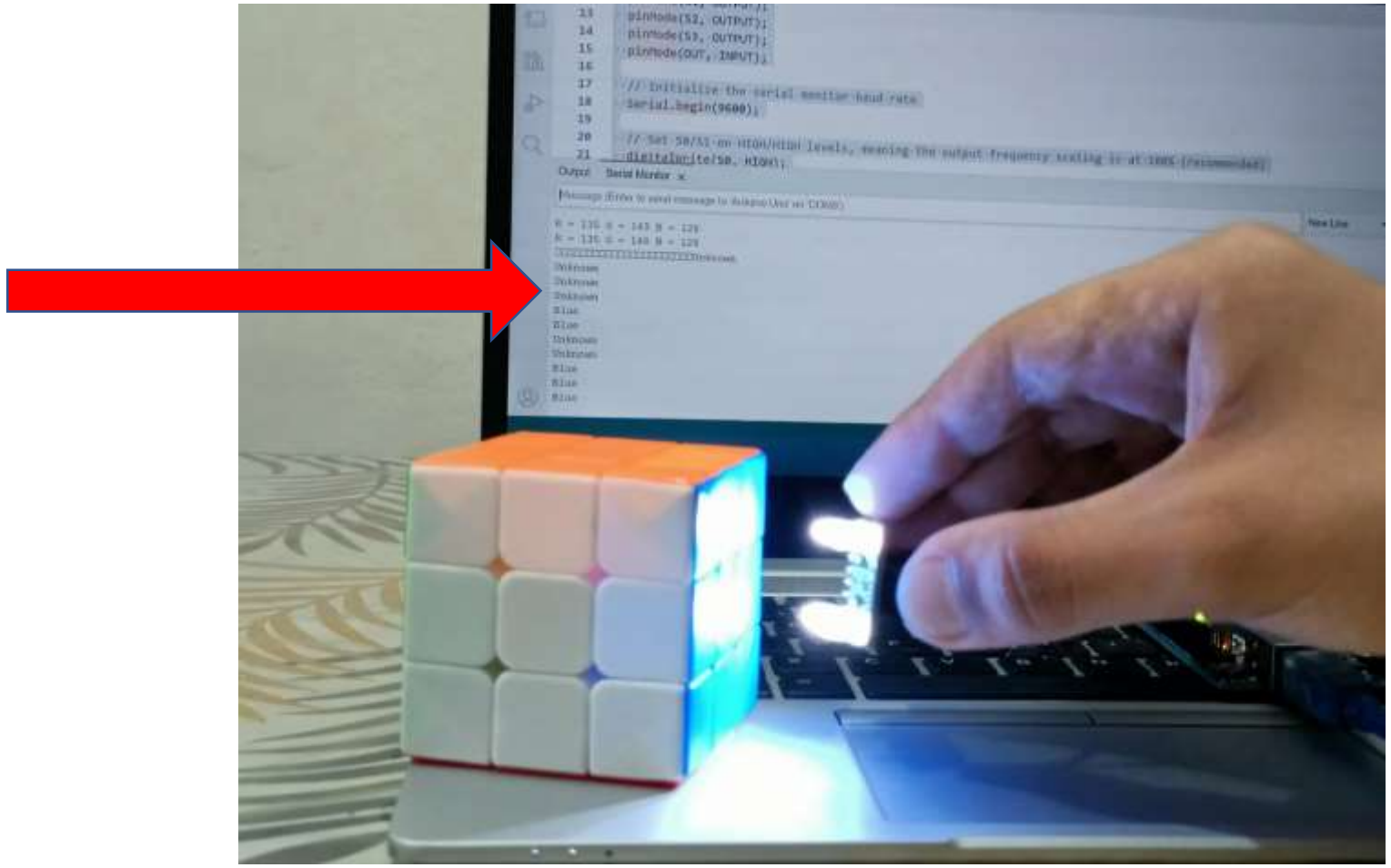


Figure 13 :détection du couleur bleu

Figure 14 :Stabilisation des Couleurs du Rubik's Cube

Le capteur de couleur ne fait pas la différence entre la couleur orange et la couleur rouge.

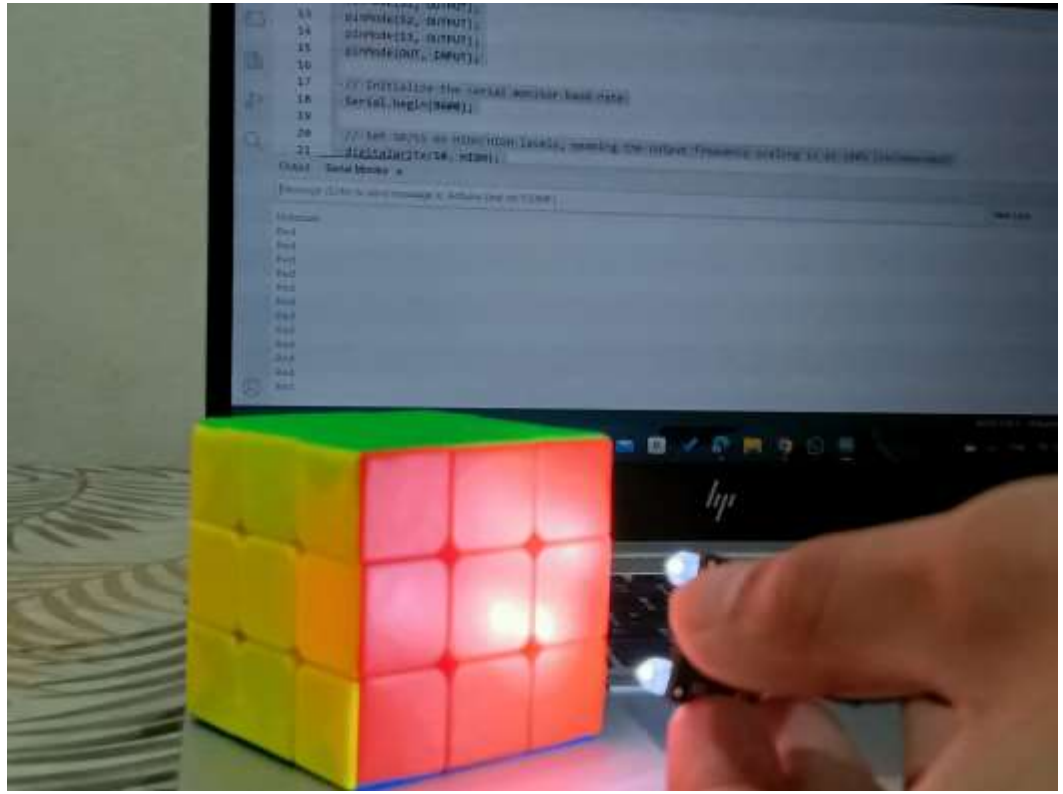


Figure 15 :détection du couleur rouge

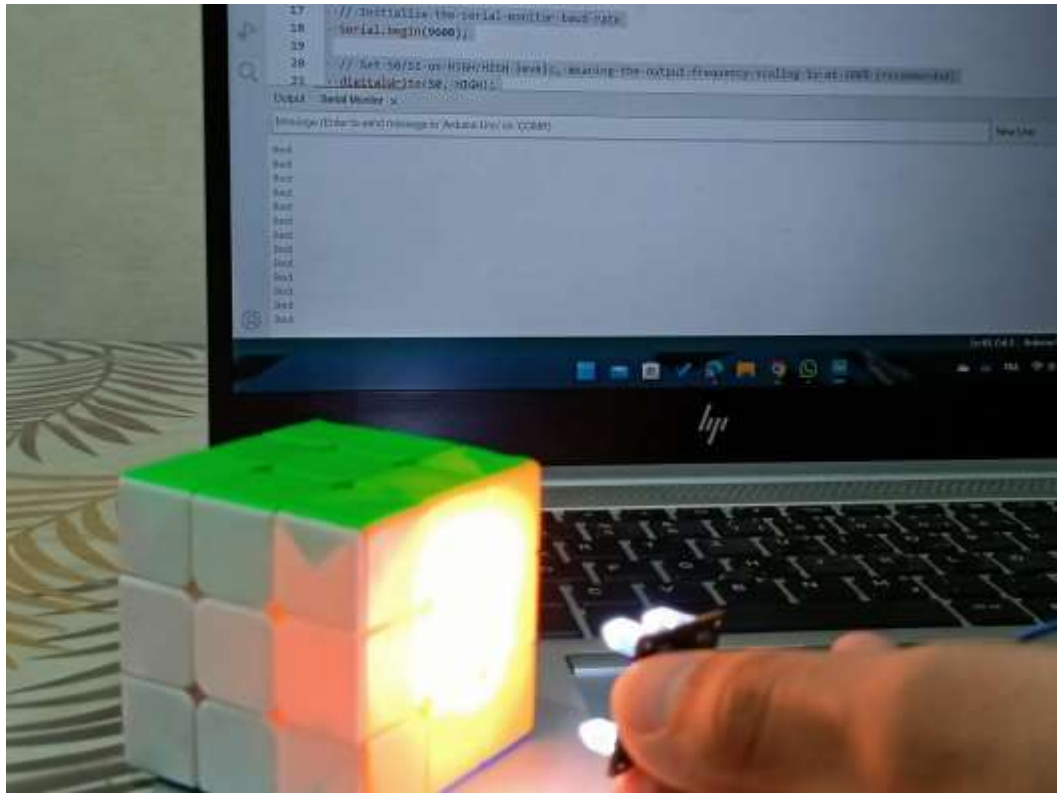


Figure 16 :détection du couleur orange

Solution pratique:

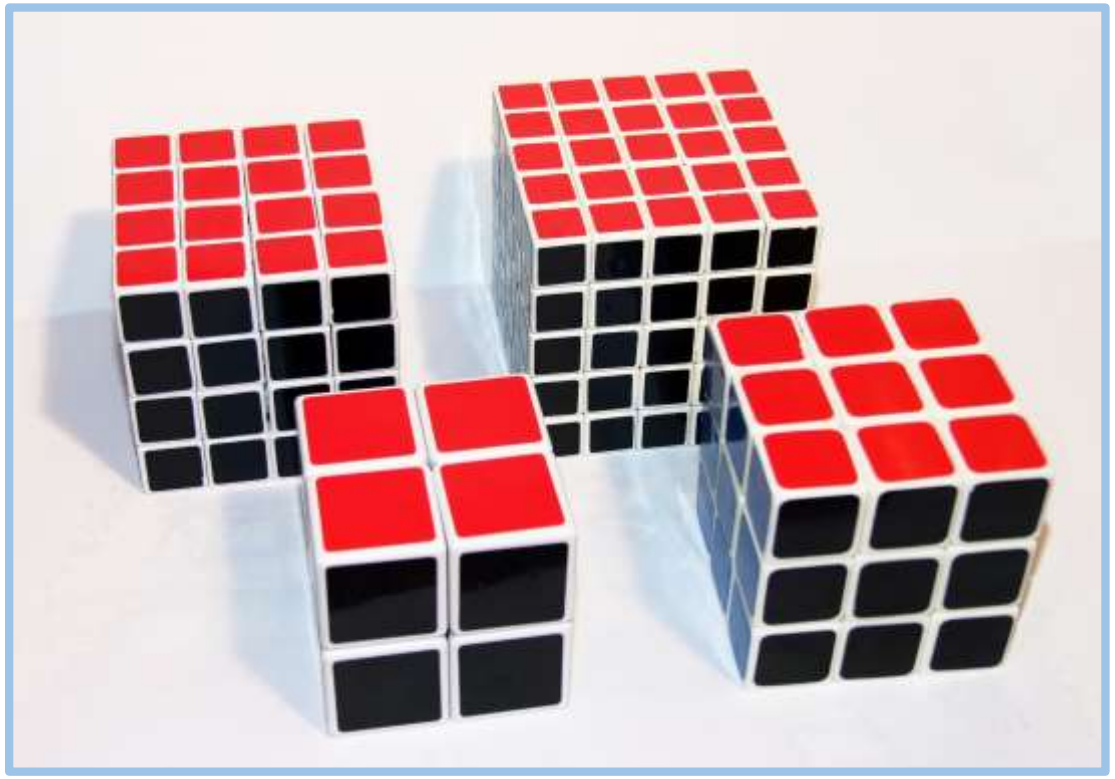


Figure 17 :Rubik cube avec couleur noir

R∈[0,15],G∈[0,15],B∈[0,15]

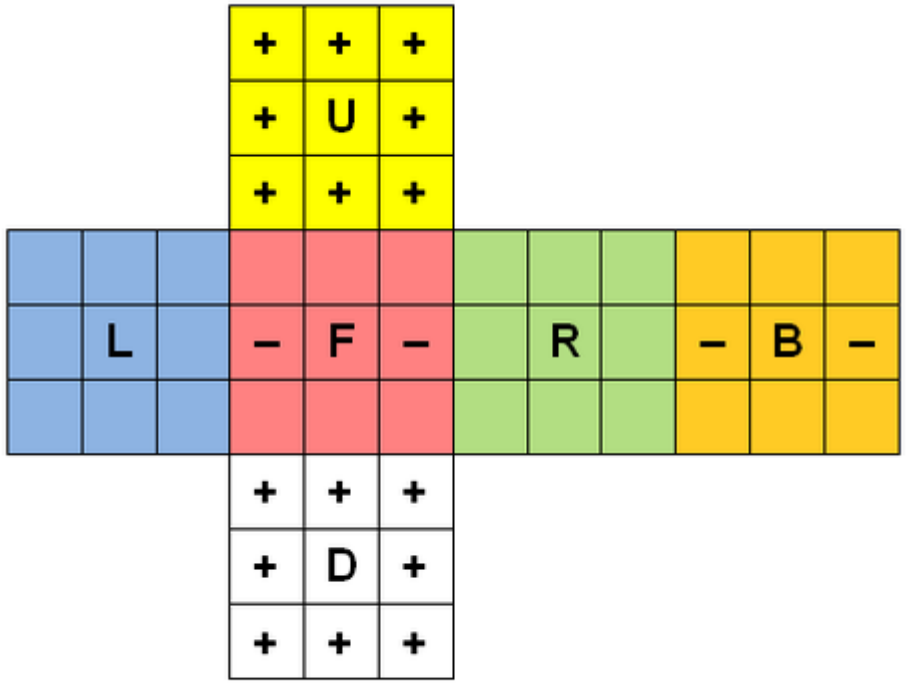
```
if (Red <= 15 && Green <= 15 && Blue <= 15)
    Serial.println("Black");
```

Figure 18 :Détection du couleur noir

l'Algorithme Kociemba:

Algorithm 1 Kociemba's Algorithm.

```
d ← 0
b ← ∞
while d < b do
    for s ∈ S^d, r(ps) = e do
        if d + d_2(ps) < b then
            Solve phase 2; report new better solution
            b = d + d_2(ps)
        end if
    end for
    d ← d + 1
end while
```



l'Algorithme Kociemba:

```python
import kociemba

cube_state = 'UUUUUUUUURRRRRRRRRFFFFFFFFFDDDDDDDDDLLLLLLLLLBBBBBBBBB'
solution = kociemba.solve(cube_state)

print("Solution:", solution)
```

La forme de solution:

"D2 R' D' F2 B D R2 D2 R' F2 D' F2 U' B2 L2 U2 D R2 U"

Calcul de couple:

$$\vec{M} = \vec{r} \times \vec{F}$$

$$|\vec{M}| = |\vec{r}| \cdot |\vec{F}| \cdot \sin(\alpha)$$

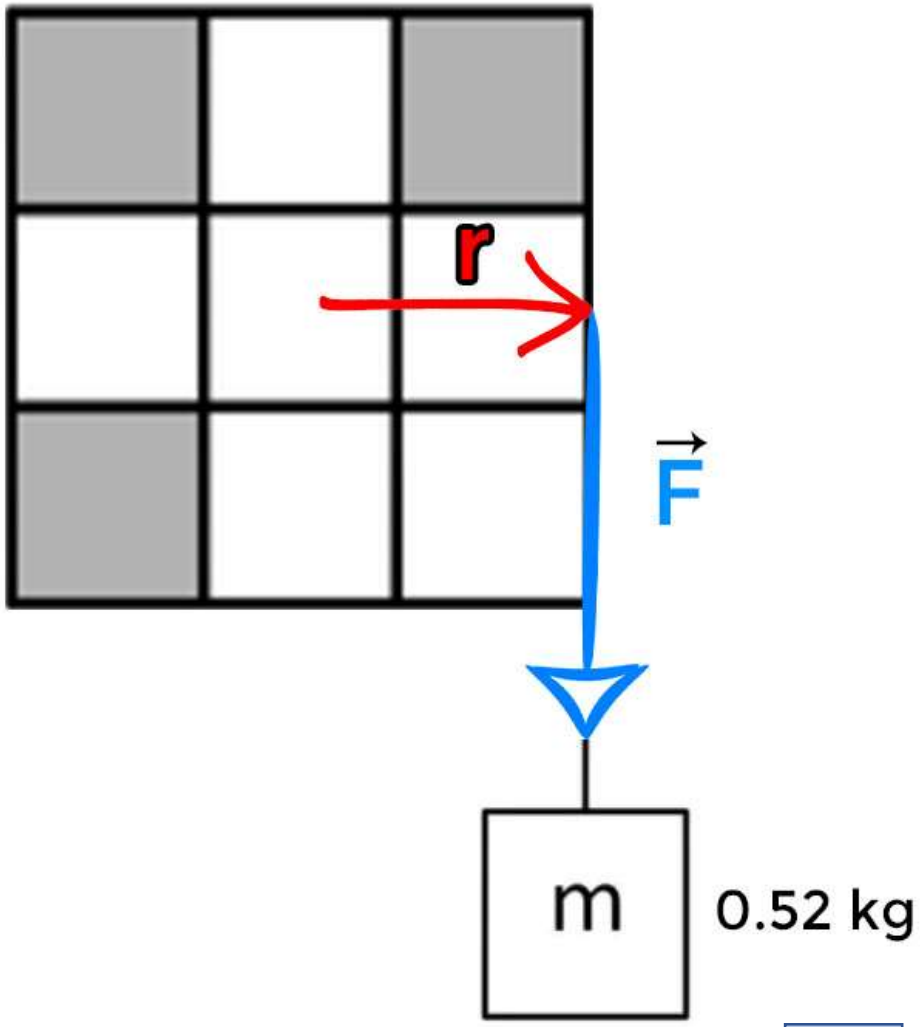
D'après le principe fondamental de la dynamique

$$F = m \cdot a$$

Dans ce cas, l'accélération est l'accélération gravitationnelle. Ainsi, le couple requis est donné par:

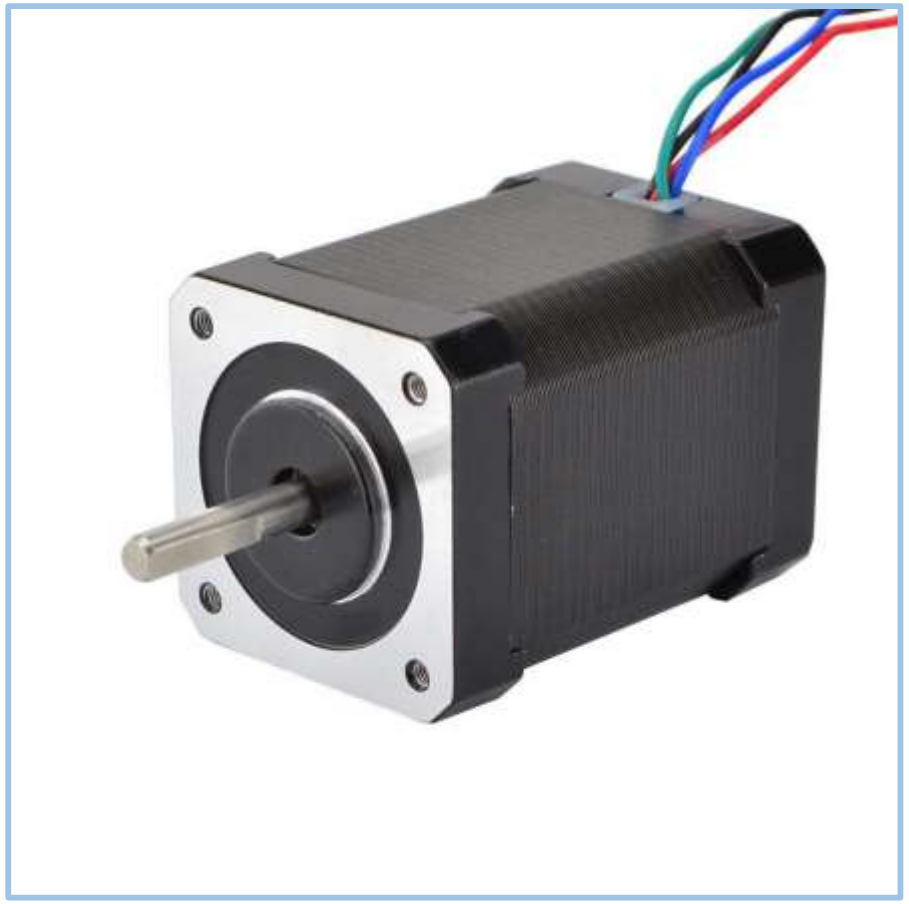
$$|\vec{M}| = r \cdot m \cdot \mathrm{g}$$

$$M = 0.0285\ m \cdot 0.53\ kg \cdot 9.81\ \frac{m}{s^2} = 0.145\ Nm$$



STEPPERONLINE Nema 17 caractéristiques:

Caractéristique	Valeur	Unité
Numéro de pièce du fabricant	17HS24-2104S	-
Type de moteur	Bipolaire	-
Angle de pas	1,8	degrés
Couple de maintien	**0.65**	**Nm**
Courant nominal/phase	2,1	A
Résistance de phase	1,6	ohms
Tension d'alimentation	3,36	V
Inductance	3,0 ± 20% (1 kHz)	mH



Conception mécanique:

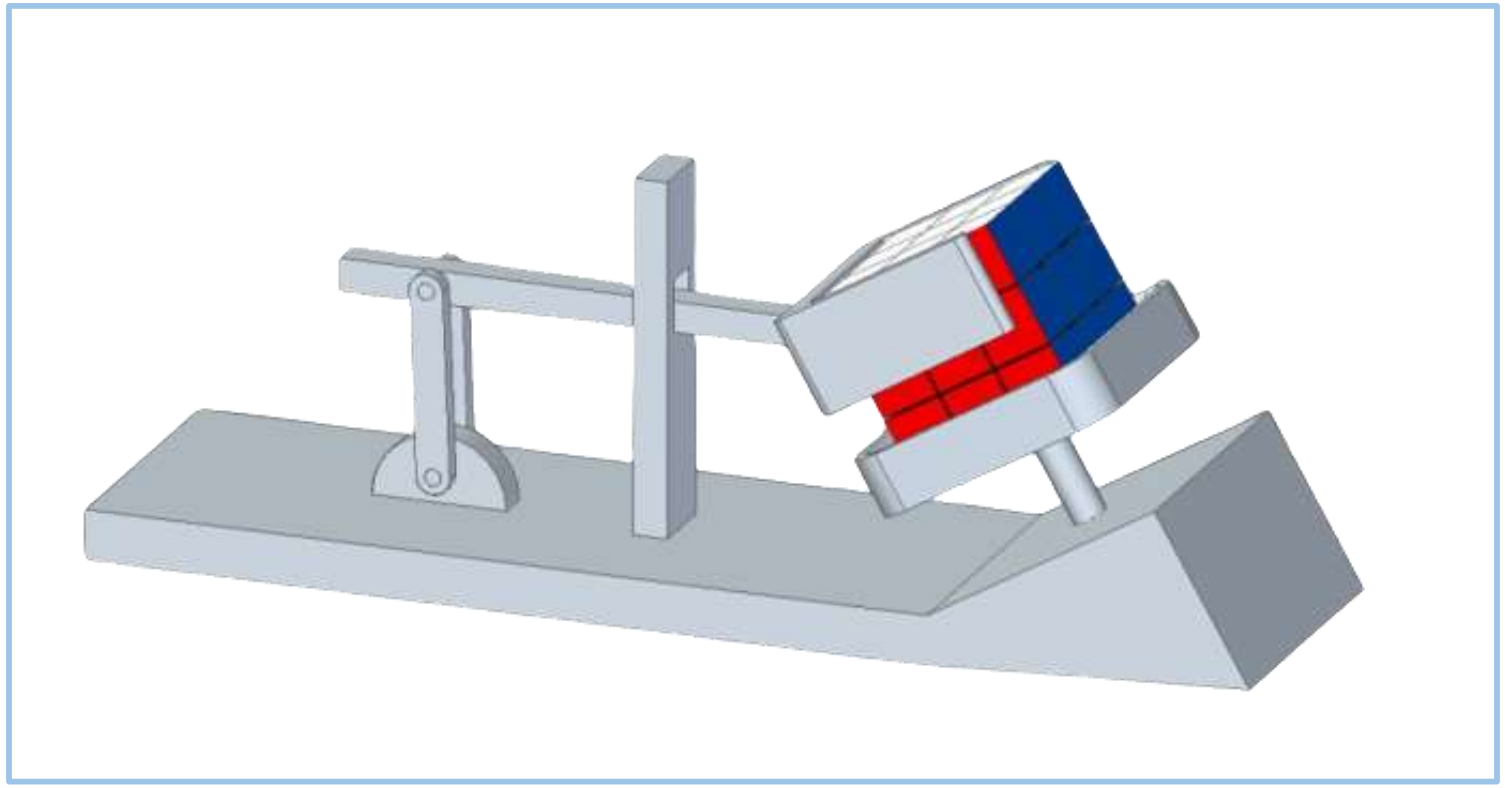
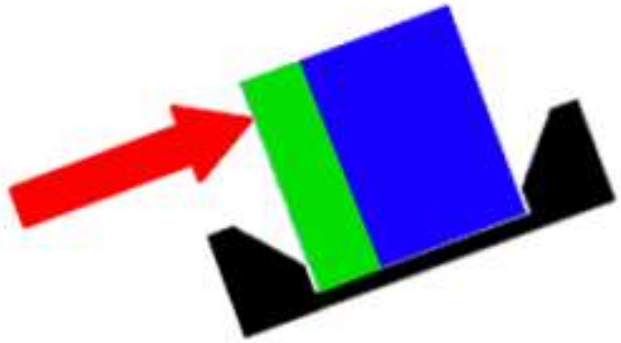
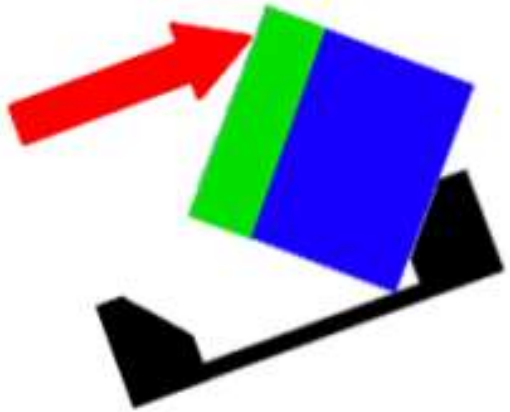


Figure 3 :Modèle 3D du robot, créé avec Solid Edge

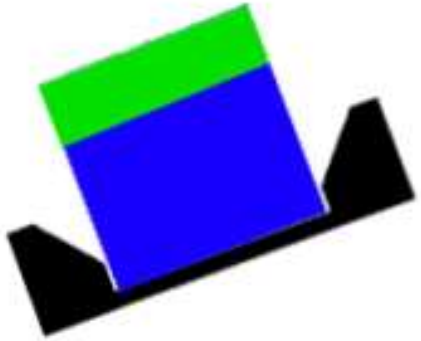
Principe de fonctionnement:



ETAPE 1



ETAPE 2



ETAPE 3

Fonctionnement du système

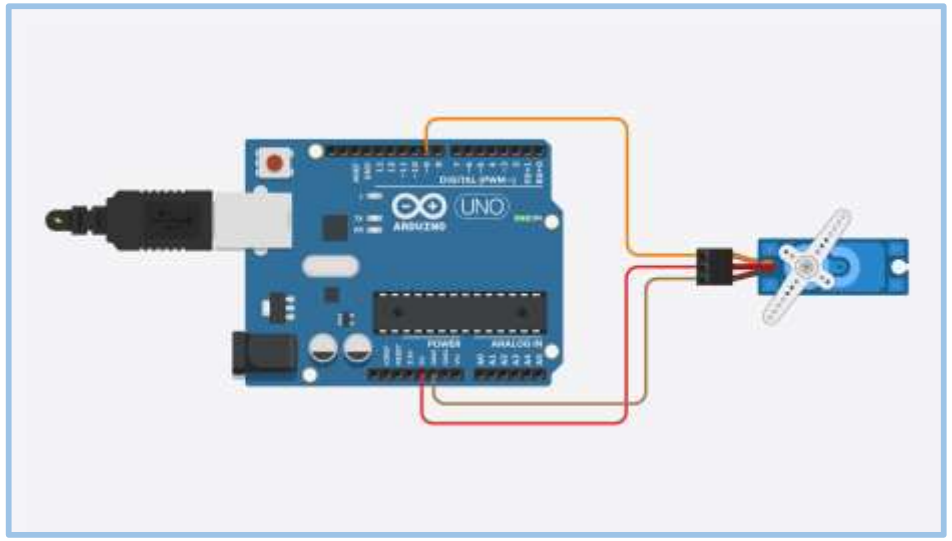
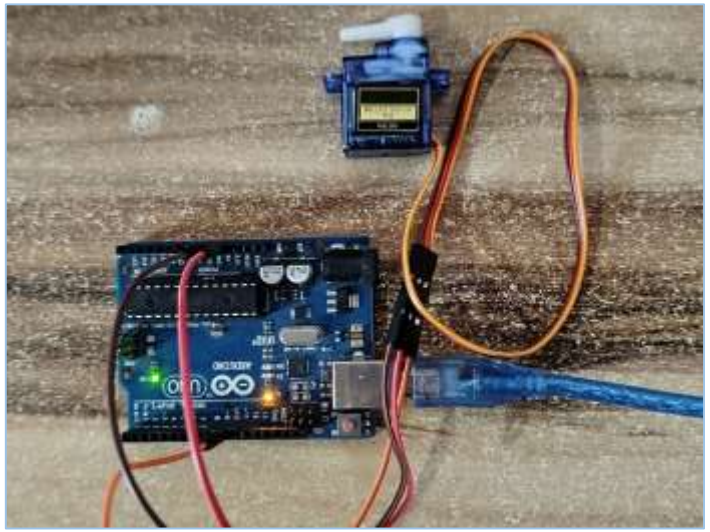


Figure 1 :Schéma éclectique du montage



```
1   #include "Servo.h"
2   Servo servo;
3   const int initialPosition = 90;
4   const int RPosition = 180;
5   const int RPrimePosition = 0;
6   void setup() {
7     servo.attach(10);
8     servo.write(initialPosition);
9     delay(1000);
10  }
11  void loop() {
12    servo.write(RPosition);
13    delay(1000);
14    servo.write(RPrimePosition);
15    delay(1000);
16    while (1);
17  }
```

Figure 1 :exemple de commande de ‘ R; R’’

CONCLUSION

Merci pour votre attention